



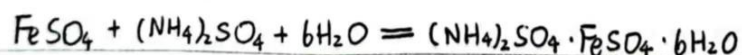
I. Objectives

- (1) Understand the concept and characteristics of double salt.
- (2) Prepare and characterize Mohr's salt.
- (3) Understand the principles and procedures of permanganate titration.

II. Principles

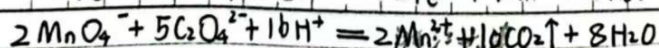
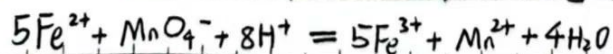
- Mohr's salt: $[(\text{NH}_4)_2\text{SO}_4 \cdot \text{FeSO}_4 \cdot 6\text{H}_2\text{O}]$, light green or blue-green crystalline compound. It is soluble in water but insoluble in several organic solvents.
It does not cake and it is less prone to oxidization, which making it a primary standard.
- Double salt: a crystalline compound that forms from a mixture of two simple salts with different cations but the same anion, and it has a distinct crystal structure compared to either of the simple salt. Solubility is lower than 2.
Therefore, Mohr's salt can be obtained through evaporation (蒸发) and cooling of its solution.

• prepare Mohr's salt:



Determine Fe: use KMnO_4 (self-catalyst and self-indicator)

When using $(\text{Na}_2\text{C}_2\text{O}_4)$ to standardize the KMnO_4 , it's important to control the temperature, acidity and titration speed.





III. Pre-lab Questions

(1). How do you prepare a standard KMnO_4 solution? Provide a brief description.

First, weight the KMnO_4 and dissolve it with water. Then, use the standard $\text{Na}_2\text{C}_2\text{O}_4$ solution to titrate the KMnO_4 solution we've prepared. Finally, calculate its solution.

(2). For the standardization of a KMnO_4 solution, what are the effects when the following conditions are higher or lower: temperature, acidity, and titration speed? Give a brief explanation.

- Temperature: i). higher: lower concentration of KMnO_4
ii). lower: higher concentration of KMnO_4 .
- Acidity: i). higher: lower concentration
ii). lower: higher concentration.
- Titration speed: i). higher: lower concentration
ii) lower: per precise concentration.

(3). When using permanganate titration to determine the iron content in the product, why should the product solution not be heated before titration?

To prevent Fe^{2+} react with O_2 in the air and turns into Fe^{3+} . Also can prevent the Mohr's salt react with KMnO_4 vice reaction.

IV. Procedures



实验名称: Preparation and Characterization of Mohr's Salt

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1. Preparation of FeSO_4 .

100 mL beaker $\xrightarrow[1.0\text{g}]{\text{iron powder}}$ $\xrightarrow[8\text{mL}]{3\text{mol/L H}_2\text{SO}_4}$ stir gently $\xrightarrow[\text{bath}]{\text{hot-water}}$ heat and stir for about 10 min
 $\xrightarrow[\text{to } 20\text{ mL}]{\text{add water}}$ stir completely $\xrightarrow{\text{suction filtration}}$ pale green and clear FeSO_4 solution.

ps. (1) If a significant amount of white or pale green solids ~~from~~ form in the beaker before suction filtration, add a little more water to dissolve them.

(2). Ensure the total volume is controlled to 30 mL, as exceeding this volume will prolong the evaporation time.

2. Preparation of $(\text{NH}_4)_2\text{SO}_4 \cdot \text{FeSO}_4 \cdot 6\text{H}_2\text{O}$.

100 mL evaporating dish $\xrightarrow[2.3\text{g}]{\text{above solution } (\text{NH}_4)_2\text{SO}_4}$ $\xrightarrow[\text{heating}]{\text{boiling water}}$ stir to dissolve completely $\xrightarrow{\text{keep concentrating}}$
form a thick crystal membrane $\xrightarrow[\text{room temperature}]{\text{slow cooling to}}$ suction filtration $\xrightarrow{\text{rinse twice with ethanol}}$ transfer on a watch glass
 $\xrightarrow[\text{for } 5\sim 10\text{ min}]{\text{dry at } 50^\circ\text{C}}$ weight the product and record.

ps: We can gradually see the crystal come out. (Refer to figures in the appendix part.)

Ps. (1). Pour about 180 mL of deionized water into a 250 mL beaker,

and heat it on a hot plate to prepare a boiling water bath.

(2). Avoid stirring the solution while concentrating to prevent the formation of small, sandy crystals.

(3). Rinse the product with ethanol twice, using 5 mL each time. (Decrease polarity of the solution).

3. Standardization of 0.02 mol/L KMnO_4 Solution.

1). Preparation KMnO_4 Solution. (0.02 mol/L).

1 L beaker $\xrightarrow[3.2\text{g}]{\text{KMnO}_4}$ $\xrightarrow[1000\text{ mL}]{\text{H}_2\text{O}}$ boil for 15 min $\rightarrow$ cool to RT $\rightarrow$ 1 L brown reagent bottle



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ii). Standardization. 加热 - 一份做一份
150 mL
Erlenmeyer flask $\xrightarrow[0.12 \sim 0.15g]{Na_2C_2O_4}$ $\xrightarrow[20 \sim 30 mL]{H_2O}$ $\xrightarrow[10 mL]{3 mol/L H_2SO_4}$ heat to 75 ~ 85 °C
→ titrate with $KMnO_4$ → from colorless to light pink
(no change within 30s) → read the final volume → repeat 2 more times.

4. Determination of the Iron Content in the Product

100 mL beaker $\xrightarrow[3.5g]{Product}$ $\xrightarrow[6 \sim 8 mL]{3 mol/L H_2SO_4}$ $\xrightarrow[30 mL]{water}$ dissolve completely $\xrightarrow{transfer}$
100 mL volumetric flask $\xrightarrow[mark]{dilute to the}$ sample solution
100 mL Erlenmeyer flask $\xrightarrow[20.00 mL]{sample solution}$ $\xrightarrow[5 mL]{3 mol/L H_2SO_4}$ titrate with 0.02 mol/L standard $KMnO_4$ solution
→ till light red solution.

V. Data Record Tables

Table 1 Preparation of Mohr's salt

Fe/g	$(NH_4)_2SO_4/g$	yield/g	Outside observation of the product
1.01	2.21	5.09	浅绿色晶体

Table 2 Standardization of 0.02 mol·L⁻¹ $KMnO_4$ Solution

	1	2	3
$m(Na_2C_2O_4) / g$	0.1227	0.1281	0.1213
$V_1(KMnO_4) / mL$	2.35	0.64	0.80
$V_2(KMnO_4) / mL$	20.68	19.76	18.80
$\Delta V(KMnO_4) / mL$	18.33	19.12	18.00

VI. Key Steps and Precautions



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1. Avoid stirring the solution while concentrating to prevent the formation of small, sandy crystals.
2. Notice the overall volume of the solution before concentrating the products. Don't keep a large amount of volume.
3. Before the titration step, don't use tap water to wash the burette. Rinse it with deionized water.
4. When titrating with the KMnO_4 , read the point where the liquid surface meets the scale on the burette.
5. When concentrating the solution, be cautious to identify the crystal coating which determines the time to stop heating.

VII. Results and Data Processing

According to the recorded data, we can get the following tables.

Table 1 Preparation of Mohr's salt

Fe/g	$(\text{NH}_4)_2\text{SO}_4/\text{g}$	yield/g	Outside observation of the product	Theoretical yield/g	Percent yield/%
1.01	2.21	5.09	浅绿色晶体	7.07	72.0

Table 2 Standardization of $0.02 \text{ mol} \cdot \text{L}^{-1} \text{KMnO}_4$ Solution

	1	2	3
$m(\text{Na}_2\text{C}_2\text{O}_4) / \text{g}$	0.1227	0.1281	0.1213
$V_1(\text{KMnO}_4) / \text{mL}$	2.35	0.64	0.80
$V_1(\text{KMnO}_4) / \text{mL}$	20.68	19.76	18.80
$\Delta V(\text{KMnO}_4) / \text{mL}$	18.33	19.12	18.00
$c(\text{KMnO}_4) / \text{mol} \cdot \text{L}^{-1}$	0.01998	0.02000	0.02012
$\bar{c}(\text{KMnO}_4) / \text{mol} \cdot \text{L}^{-1}$	0.02003		
$\bar{d}_r / \%$	0.29		

Data Treatment: In the standardization of $0.02 \text{ mol} \cdot \text{L}^{-1} \text{KMnO}_4$ Solution experiment, calculate the following treatments:

$$1: c(\text{KMnO}_4) = \frac{m(\text{Na}_2\text{C}_2\text{O}_4)}{M(\text{Na}_2\text{C}_2\text{O}_4) \times \Delta V(\text{KMnO}_4) \times 10^{-3}} \times \frac{2}{5} = 0.01998 \text{ mol} \cdot \text{L}^{-1}$$

$$2: c(\text{KMnO}_4) = \frac{m(\text{Na}_2\text{C}_2\text{O}_4)}{M(\text{Na}_2\text{C}_2\text{O}_4) \times \Delta V(\text{KMnO}_4) \times 10^{-3}} \times \frac{2}{5} = 0.02000 \text{ mol} \cdot \text{L}^{-1}$$

$$3: c(\text{KMnO}_4) = \frac{m(\text{Na}_2\text{C}_2\text{O}_4)}{M(\text{Na}_2\text{C}_2\text{O}_4) \times \Delta V(\text{KMnO}_4) \times 10^{-3}} \times \frac{2}{5} = 0.02012 \text{ mol} \cdot \text{L}^{-1}$$



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$$\bar{d}_r = \frac{1}{3} \times \frac{\sum |d_i|}{\bar{c}} \times 100\% = \frac{1}{3} \times \frac{\sum |c_i - \bar{c}|}{\bar{c}} \times 100\% = 0.29\%$$

$$\text{Relative errors: } E_{r_1} = \frac{x_1 - x_T}{x_T} \times 100\% = -0.1\%, E_{r_2} = \frac{x_2 - x_T}{x_T} \times 100\% = 0, E_{r_3} = \frac{x_3 - x_T}{x_T} \times 100\% = 0.6\%$$

$$\text{Standard deviation: } s = \sqrt{\frac{\sum_{i=1}^3 (x_i - \bar{x})^2}{2}} = 7.583 \times 10^{-5}, \text{ Class A uncertainty: } u_A = \sqrt{\frac{\sum_{i=1}^3 (x_i - \bar{x})^2}{3 \times 2}} = 0.00004$$

Therefore, the concentration of the KMnO_4 solution is $(0.02003 \pm 0.00004) \text{ mol/L}$. This indicates that the solution is very close to the 0.02 mol/L . And within a certain margin of error, we can consider its concentration is 0.02 mol/L .

VIII. Analysis, Discussion and Summary

Error analysis:

In the experiments, the yield of Mohr's salt is a little bit low and the concentration of the KMnO_4 Solution is somehow higher. Here are the possible reasons of the errors:

1. The Mohr's salt crystal is made by heating and cooling, and the precondition is the saturation solution. Therefore, it's reasonable that the yield is not 100%.
2. The cooling time maybe not so long so that the crystal's incomplete precipitation.
3. Probably the drying time is a little long, which causes the loss of water in the Mohr's salt.
4. There is some solution remaining in the containers during the pouring process.
5. In the titration process, the temperature of heating $\text{Na}_2\text{C}_2\text{O}_4$ is so high that causes it decompose.
6. While titrating, the temperature is gradually decreasing. This can also influence the usage of KMnO_4 .
7. Because the color of KMnO_4 is purplish, it's difficult to read the burette correctly.

Summary:

1. When separating the double salt, using its physical characteristics can help a lot.
2. High yield and high quality can not be gotten at the same time.
3. When analyzing the results, I tried to use more analysis methods to process my data, such as relative errors and uncertainty. I'm sure these can verify the results more persuasive.
4. To simplify the total procedures of the experiments, constrain the solution's volume.

IX. Post-lab Questions

How can all the ions in the product be identified using qualitative analysis methods? Write out all the related chemical equations.

- (1) NH_4^+ : NaOH solution and heating, use wet phenolphthalein test paper it turns to red.

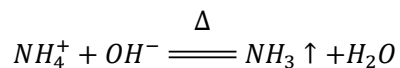


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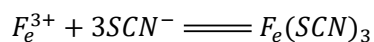
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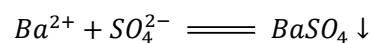
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(2) Fe^{3+} : Add KSCN solution and then it turns blood red.

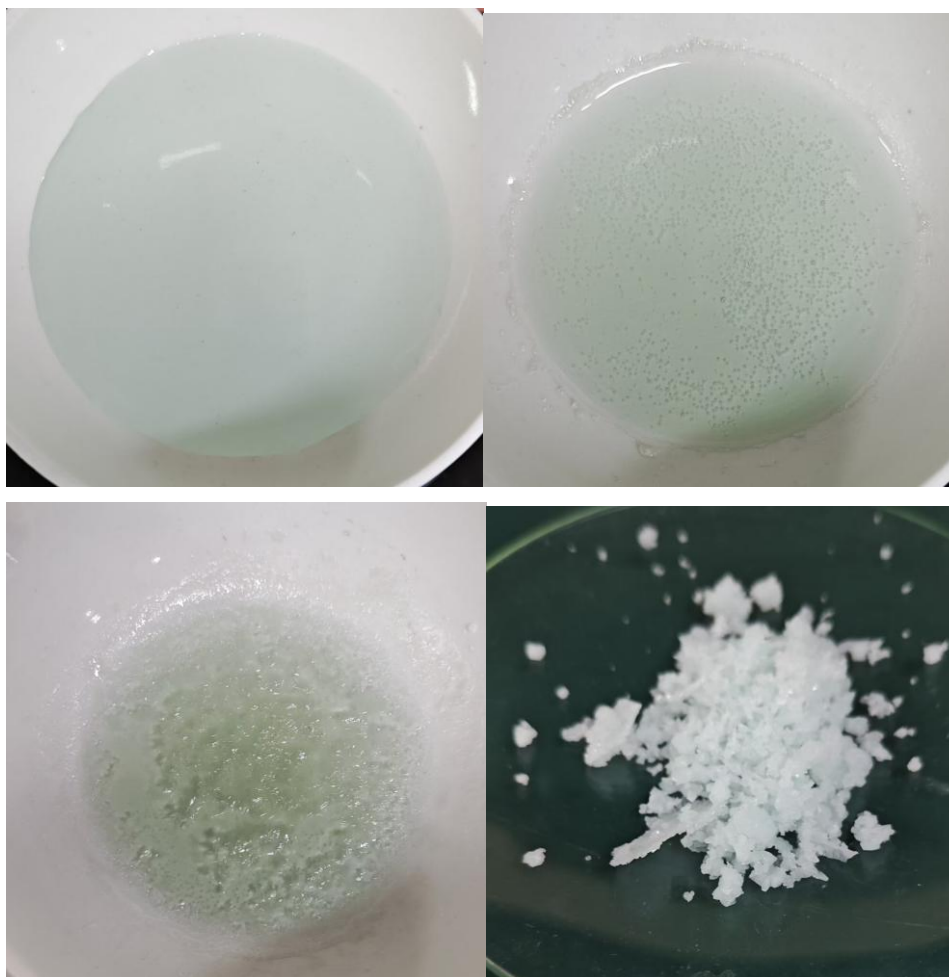


(3) SO_4^{2-} : Add $BaCl_2$ solution and it generate white precipitate. After adding HCl solution, it doesn't disappear.



X. Appendix

Here are some pictures which show the experimental phenomena during the process.



The figures shows the crystal flower, crystal film and the Mohr's salt crystal during the process of concentrating accordingly.



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评分项目	学术规范	书写工整	写作表达	数据结果和分析讨论	课前和课后思考题	总分
分值	30 分	10 分	10 分	30 分	20 分	100 分
得分/分						
评语						